DATA SCIENCE & ARTIFICIAL INTELLIGENCE



Probability and Statistics

Statistics - 1

Lecture No. 10



By- Dr. Puneet Sharma Sir

Recap of previous lecture



Topic

NORMAL DISTRIBUTION

(Part 1)

(Basic Th & G. Questions)

Topics to be Covered



Topic

NORMAL DISTRIBUTION

(Part 2)

(Remaining Theory & Extra Questions)

RECAP :-

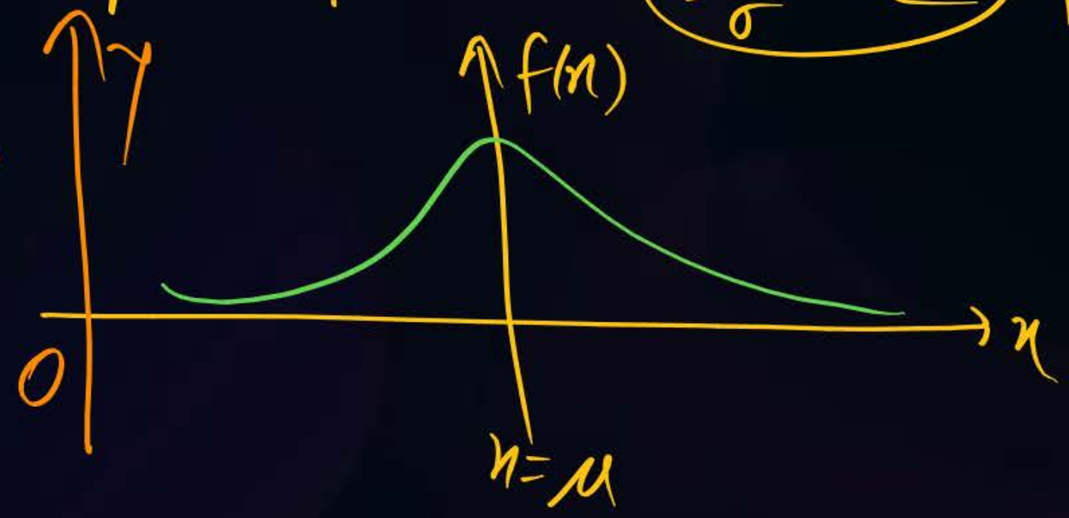
(1) $x \simeq$ N.R.V, eg Height, wt, Intelligence etc

(2) $f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left\{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right\}$ = NEND , ie $x \sim N\{\mu, \sigma^2\}$

(3) $\text{Mean}(x) = E(x) = \mu$, $\text{Var}(x) = \sigma^2$, $SD(x) = \sigma$

(4) if $\mu = 0$ & $\sigma = 1$ then $\frac{x-\mu}{\sigma} = z$ & $f(z) = \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{z^2}{2}\right\}$ = Even fun^c

(5)





ie $Z \sim N\{0, 1\}$

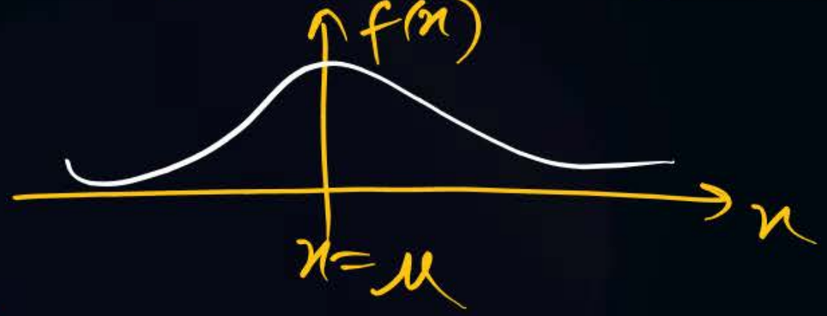
Step 1 Step 2 Step 3 Step 4

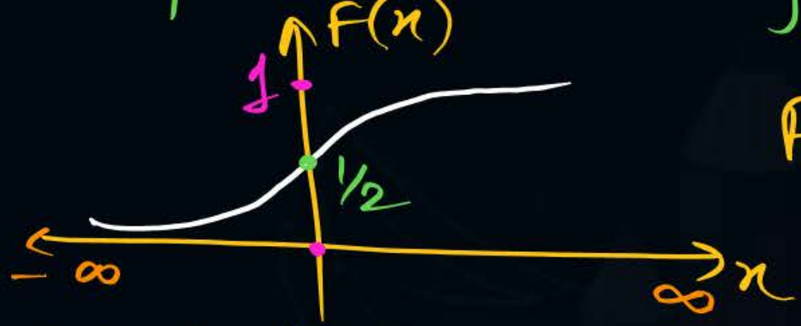
(*) $P(a<x<b) = ? = P(z_1<z<z_2)$ = use concept of symmetry = use N. Table

(*) Variance is given by width of the Bell i.e More Flat Curve ⇒ More Variance

(*) N. Curve is Bell shaped Curve with symmetry about Average.

Foreg p.d.f of N. Curve is Bell shaped while it's c.d.f is S shaped





$F(-\infty) = 0$

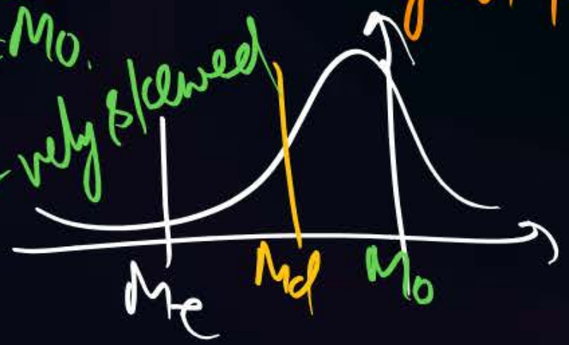
$F(\infty) = 1$

(*) N. Curve is UNIMODAL Curve with Highest Point occurs at $x=\mu$ & Highest Value is $f(\mu) = \frac{1}{\sigma\sqrt{2\pi}}$

$Me = Md = Mo$.

(*)







Note: (1) if x is N.R.V then find $F(\mu)=?$ where $f(x)$ is C.d.f at x

$$F(x)=\int_{-\infty}^{x} f(x)\,dx \Rightarrow F(\mu)=\int_{-\infty}^{\mu} f(x)\,dx=\frac{1}{2}$$

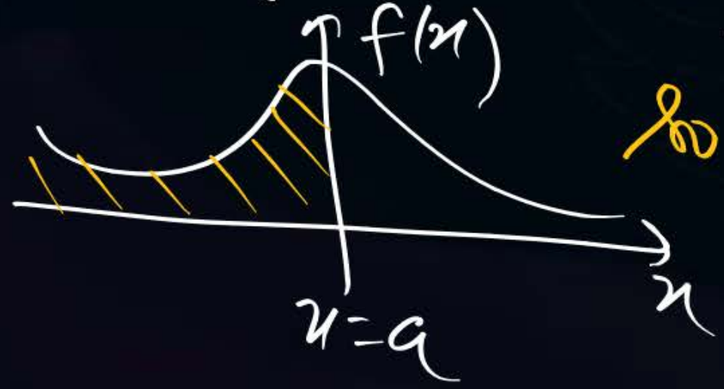


(2) 2015 If $f(x)$ is density func^n of C.R.V x defined as $f(x)=\frac{1}{b\sqrt{2\pi}}\exp\left\{\frac{-1}{2}\left(\frac{x-a}{b}\right)^2\right\}$ where a & b are statistical Attributes (parameters) then find $\int_{-\infty}^{a} f(x)\,dx=?$

Ans, $f(x)$ can be assumed as p.d.f of N. Dist. where $\mu=a$ & $\sigma=b$

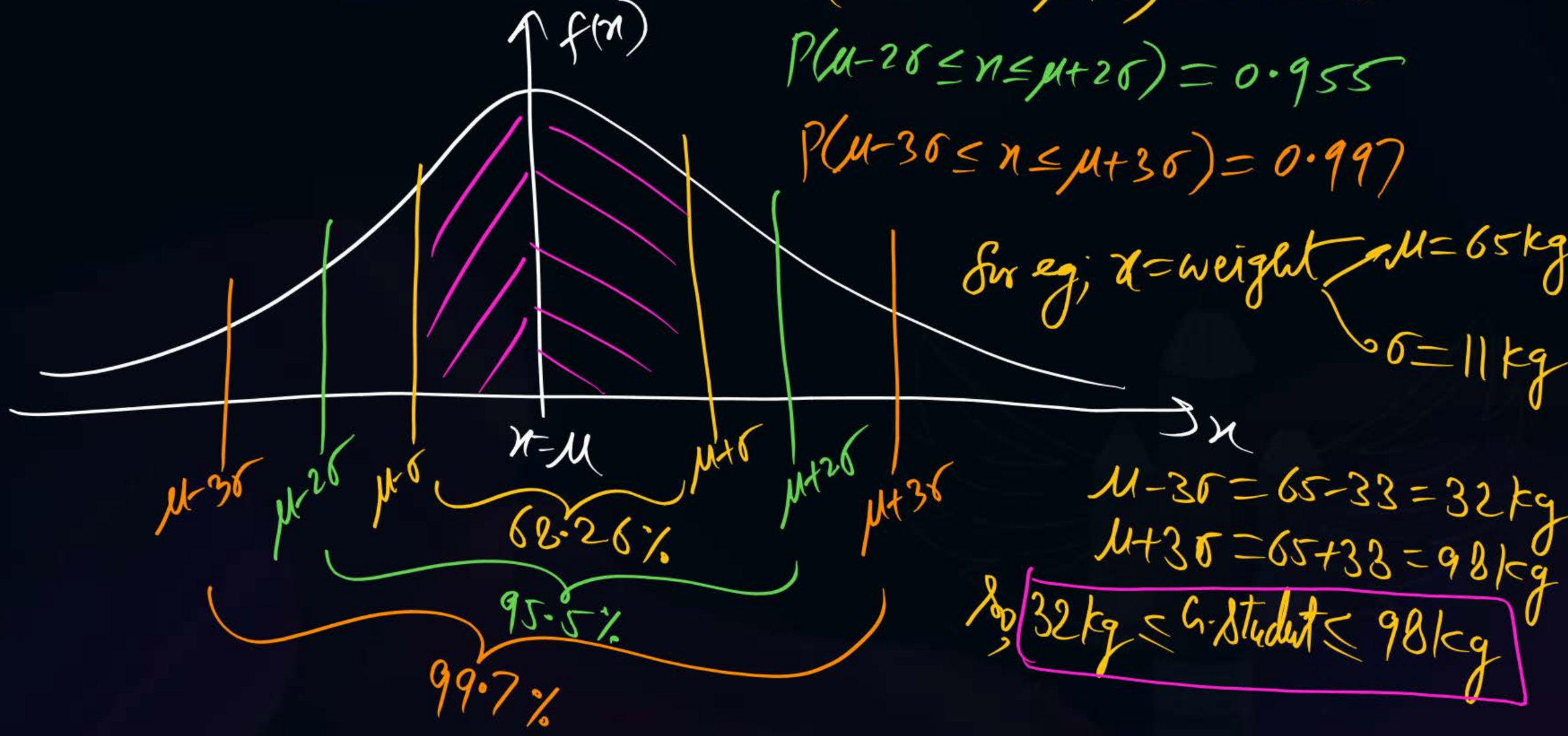
ie $x\sim N\{\mu,\sigma^2\}$

$\Rightarrow x\sim N\{a,b^2\}$



$\therefore \int_{-\infty}^{a} f(x)\,dx=?=\frac{1}{2}$

Some More Information about N. Curve →



$P(\mu-\sigma \le x \le \mu+\sigma) = 0.6826$

$P(\mu-2\sigma \le x \le \mu+2\sigma) = 0.955$

$P(\mu-3\sigma \le x \le \mu+3\sigma) = 0.997$

for eg; x = weight → $\mu = 65$ kg

$\sigma = 11$ kg

$\mu-3\sigma = 65-33 = 32$ kg

$\mu+3\sigma = 65+33 = 98$ kg

so, $32\text{kg} \le$ G. Student $\le 98\text{kg}$

(IX) At $\mu = 0, \sigma = 1, x \simeq Z$ then.

(1) $P(\mu - \sigma \le x \le \mu + \sigma) = 0.6826$

$P(0 - 1 \le Z \le 0 + 1) = 0.6826$

is $\boxed{P(-1 \le Z \le 1) = 0.6826}$

(2) $\because P(\mu - 2\sigma \le x \le \mu + 2\sigma) = 0.955$

$\Rightarrow \boxed{P(-2 \le Z \le 2) = 0.955}$

(3) $\because P(\mu - 3\sigma \le x \le \mu + 3\sigma) = 0.997$

$\Rightarrow \boxed{P(-3 \le Z \le 3) = 0.997}$

GATE

Evaluate $P(0 \le Z \le 1) = ?$

W.K. that

$P(-1 \le Z \le 1) = 0.6826$

$2P(0 \le Z \le 1) = 0.6826$

$\therefore P(0 \le Z \le 1) = 0.3413$

Ans

Q: if $x \sim N\{102, 27^2\}$ then Evaluate $P(90 \le x \le 102) = ?$

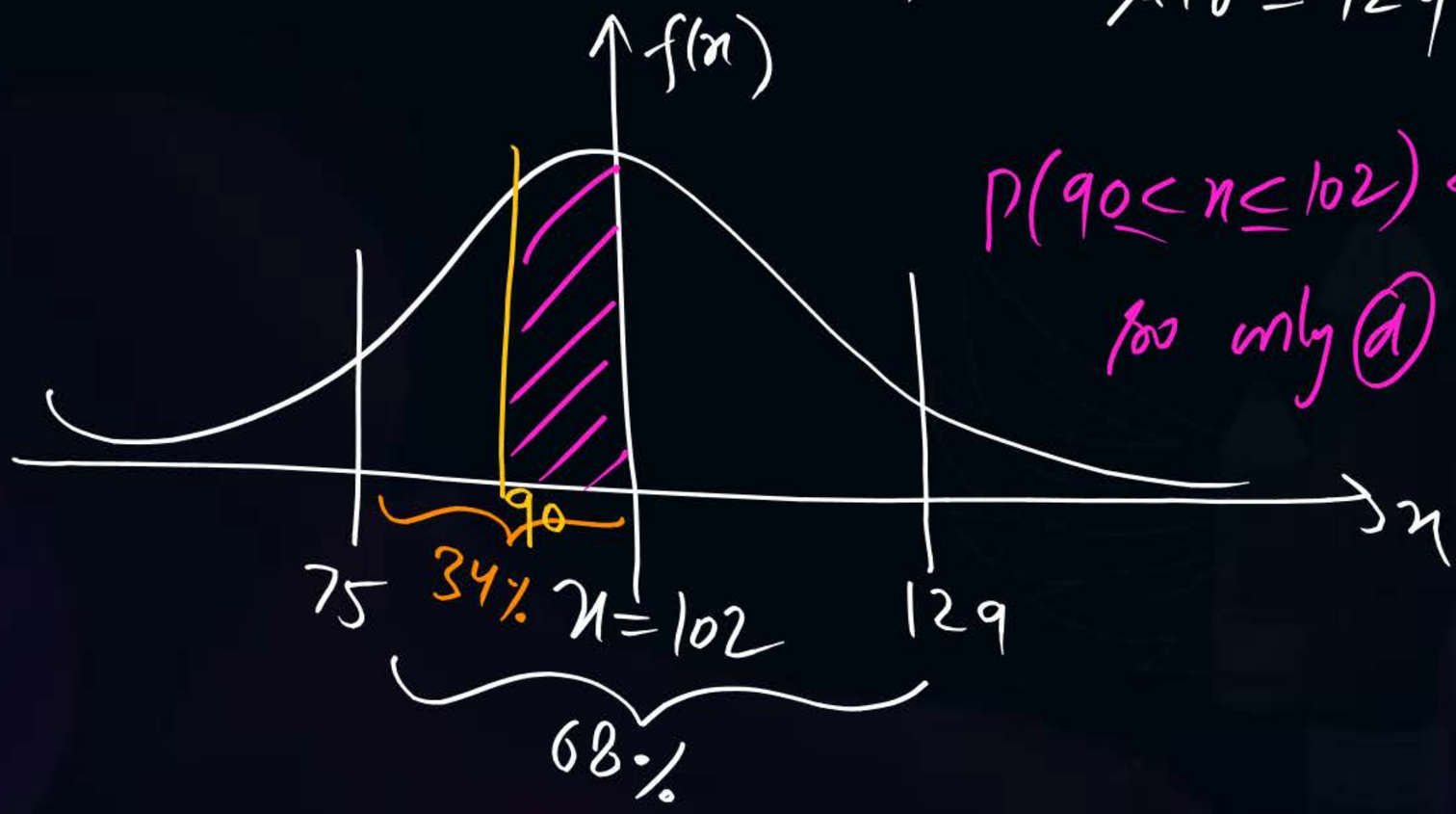
(a) 68%

(b) 50%

(c) 34%

(d) 16.7%

$x \sim N\{\mu, \sigma^2\}$ $\mu = 102$, $\sigma = 27$, $\mu - \sigma = 75$, $\mu + \sigma = 129$



$P(90 \le x \le 102) << 34\%$

so only (d) may be correct

Q: Evaluate $\int_0^\infty \frac{1}{2\sqrt{2\pi}} \exp\left\{\frac{-x^2}{8}\right\} dx = ?$ (a) 1 (b) $\frac{1}{2}$ (c) 0.25 (d) $\frac{1}{8}$

Sol: If $f(x)$ is p.d.f for N.R.V x, then

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

$$\int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx = 1$$

Put $\mu = 0, \sigma = 2$ both sides,

$$\int_{-\infty}^{\infty} \frac{1}{2\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-0}{2}\right)^2} dx = 1$$

$$\int_{-\infty}^{\infty} \frac{1}{2\sqrt{2\pi}} e^{-\frac{x^2}{8}} dx = 1$$

Even func^n

$$2\int_{0}^{\infty} \frac{1}{2\sqrt{2\pi}} e^{-\frac{x^2}{8}} dx = 1$$

$$\int_{0}^{\infty} \frac{1}{2\sqrt{2\pi}} e^{-\frac{x^2}{8}} dx = \frac{1}{2}$$

Q: If x is Gaussian Variable with Mean 1 & Variance 4.
& y is another Gaussian Variable with mean -1 & Variance unknown,
Also it is given that $P(x \le -1) = P(y \ge 2)$ then find S.D of y ?

(a) 0
(b) 2
(c) 3 ✓
(d) Data Insufficient

$x \sim N\{1, 4\}$ & $y \sim N\{-1, \sigma_y^2\}$

$$Z_x = \frac{x-\mu_x}{\sigma_x} = \frac{(-1)-1}{2} = -1, \quad \& \quad Z_y = \frac{y-\mu_y}{\sigma_y} = \frac{2-(-1)}{\sigma_y} = \frac{3}{\sigma_y}$$

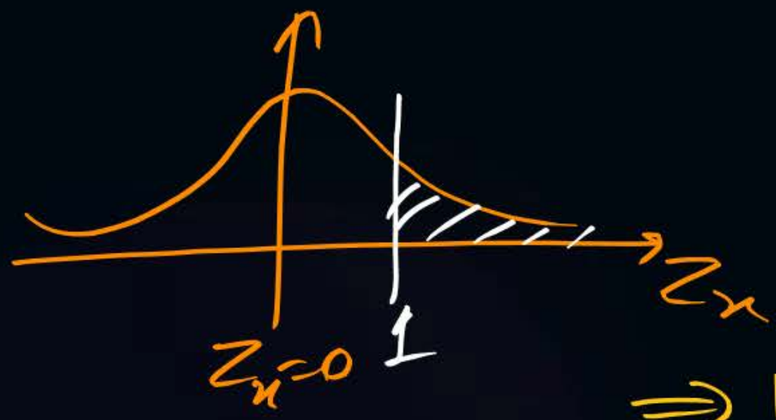
Atq, $P(x \le -1) = P(y \ge 2)$

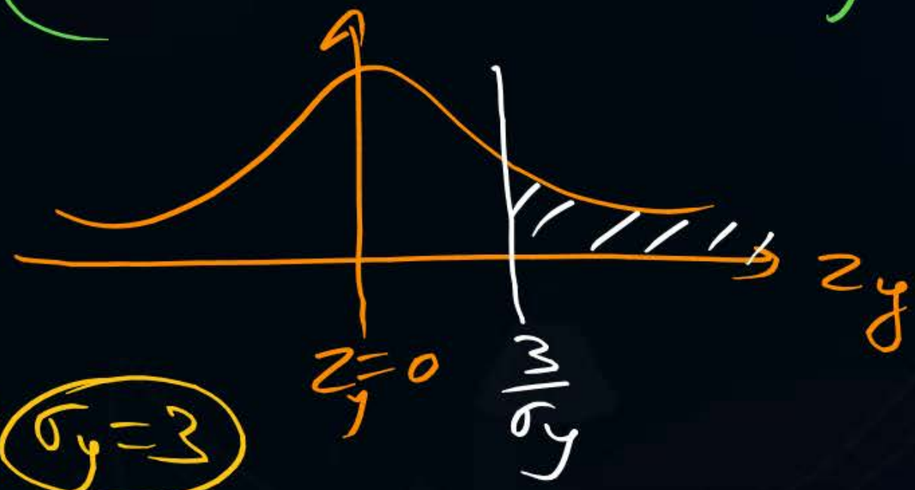
$P(Z_x \le -1) = P\left(Z_y \ge \frac{3}{\sigma_y}\right)$

$P(Z_x \ge 1) = P\left(Z_y \ge \frac{3}{\sigma_y}\right) \Rightarrow 1 = \frac{3}{\sigma_y} \Rightarrow \sigma_y = 3$

Analysis :- ① $P(Z_x \geq 1) = P\left(Z_y \geq \frac{3}{\sigma_y}\right)$

(Area under the N. Curve b/n 1 to ∞) = (Area under the N. Curve b/n $\frac{3}{\sigma_y}$ to ∞)





$\Rightarrow 1 = \frac{3}{\sigma_y} \Rightarrow \sigma_y = 3$

② wrong step.

$P(x \leq -1) = P(y \geq 2)$

$\hookrightarrow P(x \geq 1) = P(y \geq 2)$

$P(Z_x \geq 0) = P\left(y \geq \frac{3}{\sigma_y}\right)$

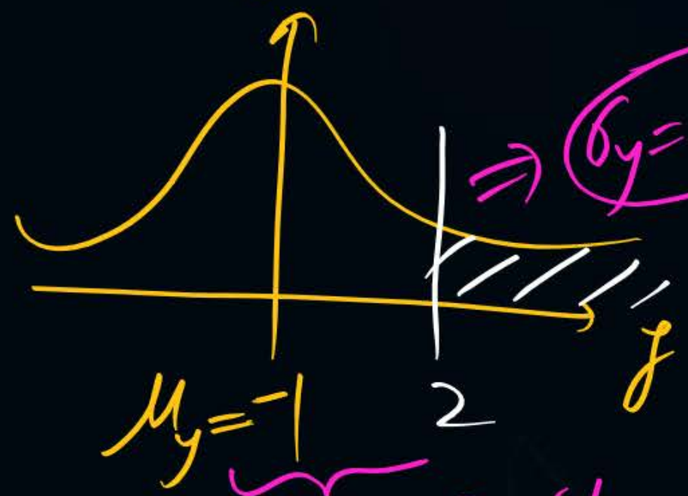
$\Rightarrow \sigma_y = \frac{3}{0} = \text{N.D.}$

Here we have changed the order of steps discussed in flowchart.

Correct Version :—

(3) $P(x \le -1) = P(y \ge 2)$





Both are same.

$P(x \ge 3) = P(y \ge 2)$

Q: If x is Zero Mean Unit Variance Gaussian Variable then

Find $E(x)$ & $E(|x|) = ?$

Sol: ATQ, $\mu_x = 0$, $\sigma_x = 1$

i.e. $\boxed{x \sim N\{0, 1\}}$

As of Now, x is in S.N. Variable form

So No Need to Convert x into z

& it's p.d.f is $f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$

= Even func^n

① $\text{Mean}(x) = E(x) = 0$ (given)

② $E(|x|) = \int_{-\infty}^{\infty} \underbrace{|x| f(x)\,dx}_{\text{Even func}^n} = 2\int_0^{\infty} |x| f(x)\,dx$

$= 2\int_0^{\infty} x \cdot \frac{1}{\sqrt{2\pi}} e^{-\left(\frac{x^2}{2}\right)} dx$

$= \sqrt{\frac{2}{\pi}} \int_0^{\infty} x \cdot e^{-\left(\frac{x^2}{2}\right)} dx$

Put $\frac{x^2}{2} = t$

$\frac{2x}{2}dx = dt$

$x\,dx = dt$

$$I = \sqrt{\frac{2}{\pi}}\int_0^\infty x\, e^{-\frac{x^2}{2}}\, dx$$

$$= \sqrt{\frac{2}{\pi}}\int_0^\infty e^{-t}\, dt = \sqrt{\frac{2}{\pi}}\left(\frac{e^{-t}}{-1}\right)_0^\infty = -\sqrt{\frac{2}{\pi}}\left(e^{-\infty} - e^{0}\right)$$

$$= -\sqrt{\frac{2}{\pi}}(0-1) = \sqrt{\frac{2}{\pi}} \simeq 0.8$$ Ans

NORMAL SUM THEOREM :-

If $x_1 \sim N\{\mu_1, \sigma_1^2\}$ & $x_2 \sim N\{\mu_2, \sigma_2^2\}$, Also x_1 & x_2 are Ind

then $(c_1 x_1 \pm c_2 x_2)$ is also a N.R.V with mean $\mu = c_1\mu_1 \pm c_2\mu_2$

& $\text{Var}(\sigma^2) = c_1^2\sigma_1^2 + c_2^2\sigma_2^2$

i.e $x \sim N\{(c_1\mu_1 + c_2\mu_2), c_1^2\sigma_1^2 + c_2^2\sigma_2^2\}$ where $x = c_1x_1 \pm c_2x_2$

Ex eg $x_1 \sim N\{\mu_1, \sigma_1^2\}$ & $x_2 \sim N\{\mu_1, \sigma_2^2\}$ x_1 & x_2 are Ind. } then $x = (x_1 + x_2)$ is also N.R.V. s.t

$x \sim N\{(\mu_1 + \mu_2), \sigma_1^2 + \sigma_2^2\}$

Eg if $x \sim N\{2,3\}$ & $y \sim N\{1,4\}$ s.t x & y are Ind then

$\mu_1=2, \sigma_1^2=3$ $\quad \mu_2=1, \sigma_2^2=4$

Find Mean and Variance of ① $2x+3y$ ② $x-y$

Sol: ① Let $X=(2x+3y)$ then By N. Sum Th $\Rightarrow \mu = 2\mu_1+3\mu_2 = 2(2)+3(1)=7$

$\sigma^2 = 4\sigma_1^2+9\sigma_2^2 = 4(3)+9(4)=48$

i.e $X \sim N\{7,48\}$

② Let $Y=(x-y)$ then By N. Sum Th, $\Rightarrow \mu = \mu_1-\mu_2 = 2-1=1$

$\sigma^2 = \sigma_1^2+\sigma_2^2 = 3+4=7$

i.e $Y \sim N\{1,7\}$

Proof of Part (1) – $x \sim N\{2,3\}$, $y \sim N\{1,4\}$ & let $X = 2x+3y$

$\mu_1 = 2, \sigma_1^2 = 3, \mu_2 = 1, \sigma_2^2 = 4$

(1) $\mu_X = E(X) = E(2x+3y) = 2E(x)+3E(y) = 2(2)+3(1) = 7$

(2) $\text{Var}(X) = \text{Var}(2x+3y) = 4\,\text{Var}(x)+9\,\text{Var}(y)+2(2)(3)\,\text{Cov}(x,y)$ (Cov → 0)

$= 4(3)+9(4) = 48$

is $X \sim N\{7, 48\}$

2013 EE

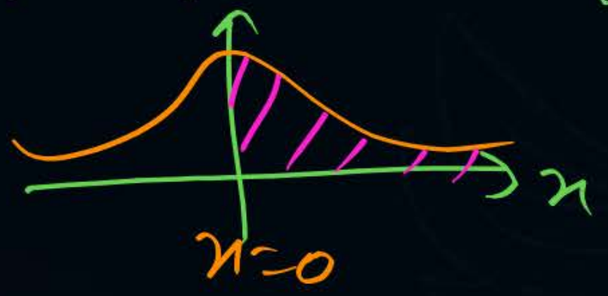
if U & V are two Ind Zero Mean Gaussian Variable having variance $\frac{1}{4}$ & $\frac{1}{9}$ resp then find $P(3V \geq 2U) = ?$

Sol: U & V are Ind, $E(U) = E(V) = 0$, $Var(U) = \frac{1}{4}$, $Var(V) = \frac{1}{9}$

$$P(3V \geq 2U) = P(3V - 2U \geq 0) = P(X \geq 0) = \frac{1}{2}$$ Ans.

Let $\boxed{X = 3V - 2U}$

∵ U & V are N.R.V. So By Normal Sum theorem X is also N.R.V



with $\mu_X = 3(0) - 2(0) = 0$ & $Var(X) = 9\left(\frac{1}{9}\right) + 4\left(\frac{1}{4}\right) = 2$ i.e. $X \sim N\{0, 2\}$

Q: Average Marks obtained by students in Maths portion of SAT exam are 529 & Variance 5732.

while in Verbal Portion, Av Marks obtained by students are 474 & Variance 6368

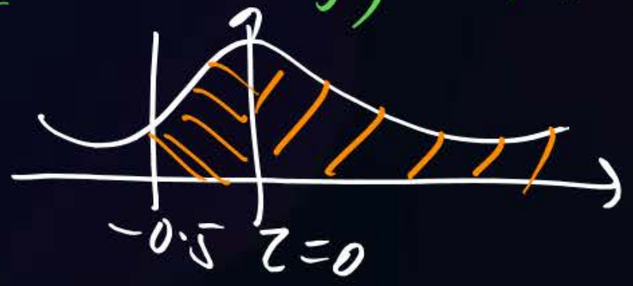
If we are selecting two students at Random then $P(x > y) = ?$

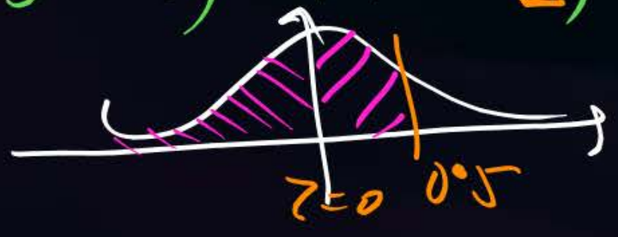
where x = Marks obtained by 1st student in Maths portion

y = " " " by 2nd student in Verbal "

& we know that area under the N. Curve upto 0.5 is 0.6915

Sol: $P(x > y) = P(x - y > 0) = P(X > 0) = P(Z_x > -0.5) = P(Z_n \le 0.5)$





$= F(0.5) = 0.6915$ Ans

Here x & y are (Ind), $\because$ Both the students are Independent.

$x \sim N\{529, 5732\}$ & $y \sim N\{474, 6368\}$

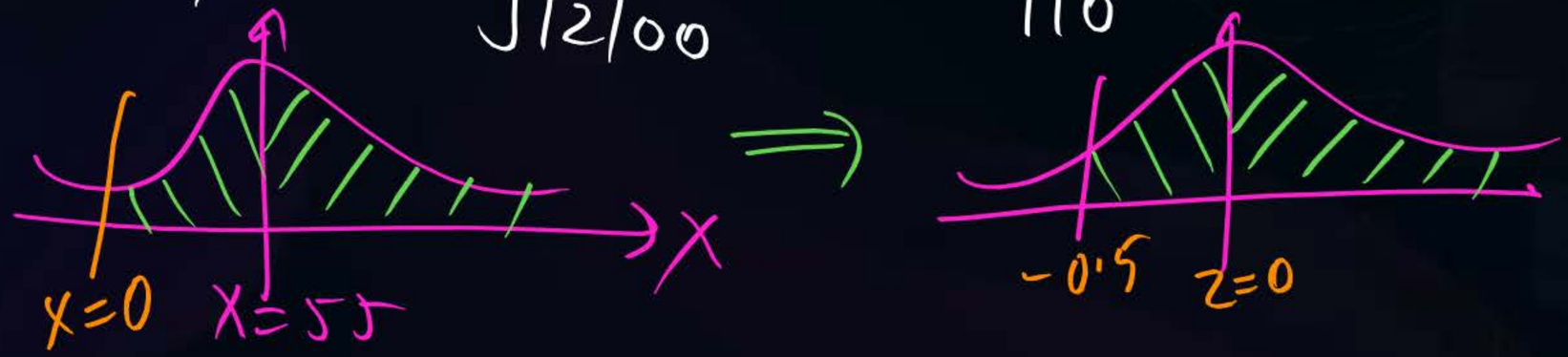
$\mu_1 = 529$, $\sigma_1^2 = 5732$, $\mu_2 = 474$, $\sigma_2^2 = 6368$

Let $X = (x - y)$

$\Rightarrow \mu_X = \mu_1 - \mu_2 = 529 - 474 = 55$

$\text{Var}(X) = \sigma_1^2 + \sigma_2^2 = 5732 + 6368 = 12100$

$$Z_X = \frac{X - \mu_X}{\sigma_X} = \frac{(0) - 55}{\sqrt{12100}} = \frac{-55}{110} = \frac{-1}{2} = -0.5$$



THANK - YOU